

NANYANG JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 2

MATHEMATICS

9758/02

Paper 2

17th September 2018

3 Hours

Additional Materials: Answer Paper
 Graph paper
 List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **8** printed pages.



NANYANG JUNIOR COLLEGE
Internal Examinations

Section A: Pure Mathematics [40 marks]

- 1** Sarah carried out a series of experiments which involved using decreasing amounts of a chemical. In the first experiment, she used 4 grams of the chemical and the amount of chemical used formed a geometric progression. In the 25th experiment, she used 1 gram of the chemical.

- (i) Find the total amount of chemical she used in the first 25 experiments. [4]
 (ii) Show that the theoretical maximum total amount of chemical she would use will not exceed 71.3 grams. [1]

Robert carried out the same series of experiments. He also used decreasing amounts of the same chemical but the amount of chemical used formed an arithmetic progression with common difference d . If the total amount of chemical that both Sarah and Robert used for the first 25 experiments were the same, and the amount of chemical Robert used for the 25th experiment was still 1 gram, find the value of d and the amount of chemical he used for the first experiment. [4]

- 2** (a) (i) Evaluate $\int \frac{2x-4}{x^2-2x+4} dx$. [3]

- (ii) Without the use of a graphic calculator, evaluate $\int_1^4 \frac{|2x-4|}{x^2-2x+4} dx$, leaving your answer in logarithmic form. [4]

- (b) Given that $\frac{d}{dx} \left(\frac{1}{\cos^2 x} \right) = \frac{2 \sin x}{\cos^3 x}$, evaluate $\int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{\cos^3 x} dx$, leaving your answer in exact form. [3]

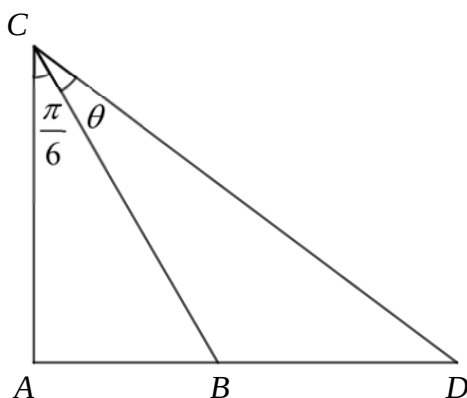
- 3** **Do not use a calculator in answering this question.**

- (i) The equation $z^3 + 4z^2 + 8z + 8 = 0$ has a root $z = -2$. Find the other two roots z_1 and z_2 where $-\pi < \arg z_2 < \arg z_1 \leq \pi$. [2]
 (ii) Find the modulus and argument of w , where $w = \frac{z_1}{z_2}$. [3]
 (iii) Find the set of positive integers n for which w^n is real, and show that, for these values of n , w^n is 1. [3]
 (iv) Express $w^{100} - (w^*)^{100}$ in the form ki , giving the exact value of k in non-trigonometrical form. [2]

- 4 (a) Hyperbolic functions are commonly used to study fluid dynamics and electromagnetic theory, where integrals with a $\sqrt{x^2+1}$ term occurs. The inverse function of the hyperbolic function $\sinh x$ is $\sinh^{-1} x$. It is given that $y = \sinh^{-1} x = \ln(x + \sqrt{x^2+1})$.

- (i) Show that $e^y \frac{d^2y}{dx^2} + e^y \left(\frac{dy}{dx}\right)^2 = (x^2+1)^{-\frac{1}{2}} - x^2(x^2+1)^{-\frac{3}{2}}$. [2]
- (ii) Hence, by further differentiation, find the first two non-zero terms of the Maclaurin's series for $\sinh^{-1} x$ in ascending powers of x . [4]

- (b) The diagram shows a right angled triangle ABC with angle $ACB = \frac{\pi}{6}$ radians. D lies on AB produced such angle $BCD = \theta$ radians.



- (i) Show that $\frac{AB}{AD} = \frac{1}{\sqrt{3} \tan\left(\frac{\pi}{6} + \theta\right)}$. [1]
- (ii) Given that θ is sufficiently small for θ^3 and higher powers of θ to be neglected, show that

$$\frac{AB}{AD} \approx 1 + a\theta + b\theta^2,$$

- where a and b are exact constants to be determined. [4]

Section B: Statistics [60 marks]

- 5 This question is about arrangements of all ten letters in the word EXCELLENCE.
- (i) Find the number of arrangements in which the letters are **not** in alphabetical order. [2]
- The letters are now arranged in a circle.
- (ii) Find the number of arrangements that can be made with all E's together and no other adjacent letters the same. [4]
- 6 A game is played using a fair six-sided die, a pawn and a simple board as shown below.

S	1	2	3	4	5	E
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Initially, the pawn is placed on square S. The game is played by throwing the die and moving the pawn in the following manner:

S 1 2 3 4 5 E 5 4 3 2 1 2 3 4 5 E.....

Thus, for example, if the first and second throw of the die gives a "5" and "4" respectively, the final position of the pawn will be on square "3".

The game will stop when the pawn stops at square E.

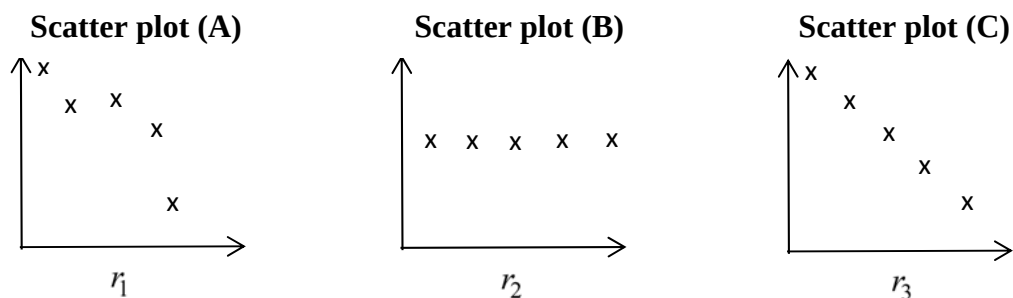
Let X be the random variable denoting the number of throws of the die required to move the pawn such that it stops at square E.

- (i) Show $P(X = 2) = \frac{5}{36}$. [1]
- (ii) Find the probability that more than two throws of the die are needed for the pawn to stop at square E given that the first throw of the die gives an even number. [3]

It is now given that for each game, a player has a maximum of 3 throws of the die and a special prize is given to any player who uses not more than two throws for the pawn to stop at square E.

- (iii) Find the probability of a player winning a special prize in at least three but not more than eight games out of ten games. [3]
- (iv) Find the least number of games needed so that the probability of winning at least a special prize is at least 0.998. [3]

- 7 (a) The following three scatter plots have product moment correlation coefficients as r_1 , r_2 and r_3 respectively.



State, with justifications, an inequality that relates r_1 , r_2 and r_3 that best describes the correlations associated with the scatter plots (A), (B) and (C). [2]

- (b) A motoring magazine published the following data on the engine capacity measured in cubic centimetres (x) and the prices in thousand dollars (y) of ten new car models.

Car Model	A	B	C	D	E	F	G	H	J	K
x	1000	1270	1750	2230	1990	600	650	1500	1450	1650
y	139	142	151.6	169.8	169.3	121.9	121.9	141.6	130.5	161.5

- (i) Plot a scatter diagram on graph paper for these values using appropriate scales for the x - and y -axes. On your diagram, indicate the car model for each point. [2]
- (ii) Find the equation of the regression line of y on x . [1]
- (iii) Calculate the product moment correlation coefficient and comment on the relationship between x and y . [2]

One particular consumer regards low price and large engine capacity as the two most important factors in choosing a car. By drawing the regression line on the scatter diagram, suggest one model which will give particularly good value for money for this consumer. Which three models would you advise the consumer not to buy? Justify your answers. [3]

- 8** The masses of manufactured links of a chain are normally distributed with mean 800 grams and standard deviation 20 grams.
- (i)** Find the probability that the mass of a randomly chosen link is more than 805 grams. [1]

To close the gate opening in a link, a locking sleeve is attached to it and it increases the mass of a link by 10%.

- (ii)** By writing down the distribution of the masses of links with locking sleeves, find the probability that the mass of a randomly chosen link with a locking sleeve is between 865.35 and 895.5 grams. [3]

Hooks with mean masses 750 grams are manufactured to attach to the links. The masses of hooks are normally distributed such that 15% of them have mass less than 735.6 grams.

- (iii)** Find the standard deviation of the masses of hooks. [2]

Five independent links with locking sleeves and a hook are packed in a wooden box with a fixed mass of 1 kilogram. The probability that the mean mass of n such wooden boxes with its contents more than 6190 grams exceeds 0.013.

- (iv)** Find the greatest value of n , stating the parameters of any distribution that you use. [4]

Question 9 is printed on the next page

- 9 During a Mathematics lesson, Ms Kim wanted her pupils to build a model. She carried two indistinguishable boxes of bricks into class. The bricks are identical and indistinguishable except for colours. The number of coloured bricks found in each box is as follows.

Colour of Bricks	Box 1	Box 2
Blue	2	4
Red	3	3
Yellow	5	3

A pupil, Donald, has to draw two bricks from these boxes randomly to build a model. He draws a brick randomly from one of the boxes. The brick is not replaced. He then draws a second brick randomly from one of the boxes.

- (i) Show the probability that Donald does not draw any yellow brick is $\frac{25}{72}$. [2]
- (ii) Find the probability that Donald draws two yellow bricks. [3]
- (iii) Tabulate the probability distribution table for the number of yellow bricks drawn and find the expected number of yellow bricks drawn by Donald. [2]

Ms Kim decided to combine the two boxes of bricks into a single box.

- (iv) Donald draws 5 bricks **with replacement** from the single box. Find the probability that he draws less than two yellow bricks. [2]
- (v) Donald wanted to build the model with a yellow brick. He randomly draws a brick from the single box one at a time **with replacement** until he gets a yellow brick. Find the probability that it will not take him more than nine draws to get a yellow brick. [3]

Question 10 is printed on the next page

- 10 The distances thrown, x metres, by a discus thrower have been recorded over a long period. The results for a random sample of 60 throws are summarised by $\sum(x - 65) = 120$ and $\sum(x - 65)^2 = 3810$.

- (i) State what it means for a sample to be random in this context. [1]
- (ii) Calculate the unbiased estimates of the population mean and variance of the distances thrown by the discus thrower. [2]

An analysis of all the data collected over the long period gives a mean of 68 metres and a standard deviation of 7.5 metres.

The Sports Council wishes to study if a new technique will improve the mean distance thrown by this thrower. Using this new technique, the results of a random sample of n throws give a mean of $\bar{x}$ metres. A test is carried out, at the 2% level of significance, to determine whether the mean distance thrown by the thrower has improved. You may assume that the distances thrown by the thrower follows a normal distribution.

- (iii) State appropriate hypotheses for the test, defining any symbols you use. [1]
- (iv) Given that $n = 30$, find the range of values of $\bar{x}$ for which the result of the test would be to reject the null hypothesis. [3]
- (v) It is given instead that $\bar{x} = 70.1$ and the result of the test is that the null hypothesis is not rejected. Obtain an inequality involving n , and hence find the set of values that n can take. [3]

A sports magazine published a claim about the mean distance thrown using this new technique. It is assumed that the distances of the discuses thrown follow a normal distribution with known population variance.

Two discus thrower coaches, Ang and Tan, decide to each conduct a hypothesis test, at the 5% level of significance, to determine whether the magazine has **overstated** the mean distance thrown using this new technique. Ang obtained a sample with mean $\bar{x}$ and concluded that there is sufficient evidence, at the 5% level of significance, that the magazine has overstated the mean distance thrown. Tan took a different sample with sample size **four times** that of Ang's and obtained the same sample mean.

Will Tan's test yield the same conclusion? Justify your answer. [2]